

ARKADELPHIA, AR, USA

WMO: 720394

Lat: 34.100N

Lon: 93.066W

Elev: 183

StdP: 14.60

Time Zone: -6.00 (NAC)

Period: 07-19

WBAN: 00428

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.4	24.5	9.1	8.8	27.9	14.0	11.3	31.1	19.0	55.2	17.2	54.0	3.3	330	0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.1	99.9	78.8	97.3	78.6	94.6	78.4	82.2	92.9	81.2	91.8	80.3	90.7	5.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.5	154.2	85.9	78.9	150.9	85.6	77.7	144.8	85.1	45.8	92.6	44.8	91.4	43.8	91.0	86.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.6	14.0	12.0	14.4	101.6	4.1	4.1	11.4	104.6	9.0	107.0	6.7	109.3	3.7	112.3
			13.5	83.9	4.1	1.2	10.5	84.7	8.1	85.4	5.8	86.0	2.8	86.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.4	41.9	46.7	55.8	63.5	71.6	80.2	82.0	81.7	75.8	63.5	52.2	45.3	(6)
	DBStd	16.10	9.41	10.09	9.03	6.95	6.22	4.13	4.35	4.62	5.82	8.21	8.88	8.84	(7)
	HDD50	789	282	168	47	4	0	0	0	0	0	5	79	204	(8)
	HDD65	2798	717	514	306	107	20	0	0	0	2	128	393	611	(9)
	CDD50	5697	31	77	227	410	668	907	992	984	774	424	144	59	(10)
	CDD65	2230	1	3	21	62	223	457	527	519	326	82	8	1	(11)
	CDH74	23501	2	19	196	602	1950	5002	5899	5650	3166	921	83	11	(12)
	CDH80	11438	0	0	26	108	691	2572	3137	3019	1540	334	11	0	(13)
	WSAvg	4.1	4.9	5.4	5.5	5.1	4.1	3.8	3.0	2.7	2.8	3.3	3.9	4.4	(14)
(15) Precipitation	PrecAvg	55.8	4.4	4.1	5.3	5.1	6.1	4.2	4.2	2.8	4.3	5.5	4.6	5.1	(15)
	PrecMax	87.7	9.7	8.8	11.9	10.7	15.7	12.4	13.0	9.4	11.4	16.1	11.5	10.8	(16)
	PrecMin	34.6	0.2	0.7	1.6	2.2	2.2	1.0	0.2	0.4	1.2	0.1	1.1	0.6	(17)
	PrecStd	11.0	2.6	1.8	2.7	2.4	2.9	2.8	3.0	1.8	2.4	3.8	2.4	2.4	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.9	76.8	82.6	85.9	91.4	100.3	101.9	105.9	99.7	91.4	81.7	75.3	(19)
		MCWB	61.7	65.3	68.0	70.7	76.1	77.5	79.1	80.0	76.2	72.2	69.8	69.9	(20)
	2%	DB	67.6	72.4	79.5	82.5	89.6	97.1	99.5	100.3	96.1	87.8	75.4	70.2	(21)
		MCWB	59.5	64.2	65.6	68.3	75.0	78.5	79.3	79.5	75.9	71.8	64.9	65.6	(22)
	5%	DB	63.3	68.7	75.1	80.9	87.0	94.1	97.0	97.1	92.4	83.7	72.1	65.8	(23)
		MCWB	56.5	61.6	63.5	67.7	74.0	78.1	79.6	78.7	75.8	70.0	64.8	61.0	(24)
	10%	DB	58.4	64.3	71.8	77.8	84.1	91.8	93.4	93.3	89.4	79.4	67.9	61.2	(25)
		MCWB	51.4	58.6	62.1	66.5	72.9	77.5	79.2	78.6	75.3	68.7	62.2	56.7	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.3	68.6	70.7	75.1	79.5	82.5	83.3	83.5	80.5	78.1	72.7	71.0	(27)
		MCDB	70.5	73.1	79.7	81.4	88.3	92.5	94.1	95.2	90.4	86.8	77.7	71.9	(28)
	2%	WB	62.6	66.2	68.6	72.4	77.6	81.2	82.0	82.3	79.1	75.9	68.8	66.6	(29)
		MCDB	65.7	70.4	76.6	79.1	86.0	91.3	92.7	93.4	89.1	83.8	73.4	69.5	(30)
	5%	WB	57.9	63.4	66.3	70.5	76.1	80.0	81.1	81.3	78.0	73.3	66.2	62.3	(31)
		MCDB	61.0	66.8	72.6	77.0	83.6	90.5	91.7	92.0	88.0	79.1	70.4	64.3	(32)
	10%	WB	52.8	59.7	63.9	68.7	74.8	78.8	80.2	80.3	76.8	70.5	62.9	57.2	(33)
		MCDB	56.9	63.7	69.4	74.8	81.6	89.1	90.8	90.6	86.2	75.7	66.8	60.2	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	21.4	20.5	22.7	22.6	20.6	21.5	21.1	21.6	23.4	25.4	23.5	19.8	(35)
		MCDBR	29.0	25.9	27.1	26.7	23.9	24.9	26.0	27.4	28.3	30.8	27.9	26.2	(36)
	5% WB	MCWBR	20.4	18.1	15.3	13.9	10.9	8.9	9.0	9.3	10.8	15.3	18.2	20.8	(37)
		MCWBR	23.3	20.8	22.1	21.3	20.5	21.6	22.3	23.4	22.7	23.3	22.1	22.9	(38)
(40) Clear-Sky Solar Irradiance	taub		0.329	0.334	0.366	0.399	0.434	0.466	0.488	0.484	0.414	0.364	0.341	0.319	(40)
		taud	2.491	2.476	2.409	2.317	2.281	2.219	2.187	2.188	2.361	2.477	2.494	2.529	(41)
	Ebn at Noon		279	290	288	282	272	262	255	254	269	276	272	276	(42)
		Edh at Noon	27	30	35	41	43	45	46	46	37	30	27	25	(43)
(44) All-Sky Solar Radiation	RadAvg		809	935	1316	1646	1800	2010	1966	1820	1561	1225	875	708	(44)
	RadStd		75	142	110	104	154	172	122	145	131	170	102	83	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (25 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page